Taylor's theorem for a function of two variables

For a function of a single variable, we have studied the Taylor's theorem, which has many applications, especially in the development of algorithms in numerical analysis. In this section, we will study the Taylor's expansion of functions of two variables.

Theorem:

If $f(x, y)$ possesses continuous partial derivatives of nth order in any neighbourhood of a point (a, b) and if $(a + h, b + k)$ is a point in this neighbourhood, then there exists at least $\theta \in (0, 1)$ such that

$$f(a+h, b+k) = f(a,b) + \left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right) f(a,b) + \frac{1}{2!}\left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)^2 f(a,b) + \cdots$$
$$+ \frac{1}{(n-1)!}\left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)^{n-1} f(a,b) + \frac{1}{n!}\left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)^{n} f(a+\theta h, b+\theta k),$$

$$0 < \theta < 1.$$

Proof: Let (x, y) be any point in the neighbourhood of the point (a, b). Then we can write $x = a + ht, y = b + kt$.

$\therefore f(x, y) = f(a + ht, b + kt) = g(t)$ is a function of a single variable t. By the assumptions of the theorem, $g(t)$ possesses continuous derivatives of order n in a neighbourhood of the origin.

Hence, $g(t)$ can be expanded by a Maclaurin's series

$$g(t) = g(0) + tg'(0) + \frac{t^2}{2!}g''(0) + \cdots + \frac{t^{n-1}}{(n-1)!}g^{(n-1)}(0) + \frac{t^n}{n!}g^{(n)}(\theta t), \qquad 0 < \theta < 1$$

For $t = 1$,

$$g(1) = g(0) + g'(0) + \frac{1}{2!}g''(0) + \cdots + \frac{1}{(n-1)!}g^{(n-1)}(0) + \frac{1}{n!}g^{(n)}(\theta), \qquad 0 < \theta < 1$$

... **(1)**

Now, $g(0) = f(a, b)$

$g(t) = f(a + ht, b + kt)$

$\Rightarrow$ By total differentiation formula

$$g'(t) = \frac{d}{dt}\big(f(a+ht, b+kt)\big) = \frac{d}{dt}f(x,y)$$

$$= \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}, \qquad x = a + ht, y = b + kt$$

$$= \frac{\partial f}{\partial x}h + \frac{\partial f}{\partial y}k = \left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)f$$

$\therefore g'(0) = \left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right) f(a,b)$

$\therefore \frac{d}{dt} = h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}$

$$g''(t) = \frac{d}{dt}g'(t) = \left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)\left(g'(t)\right)$$

$$= \left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)\left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right) f$$

$$= \left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)^2 f$$

Similarly, we can show that

$$g^{(k)}(t) = \left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)^k f, \qquad k = 1, 2, \dots n.$$

Substituting in (1), we get

$$g(1) = f(a+h, b+k)$$
$$= f(a,b) + \left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right) f(a,b) + \frac{1}{2!}\left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)^2 f(a,b) + \cdots$$
$$+ \frac{1}{(n-1)!}\left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)^{n-1} f(a,b) + \frac{1}{n!}\left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)^n f(a+\theta h, b+\theta h)$$

$$0 < \theta < 1.$$

Note

1. If we consider $a = a + h, y = b + k$, then $h = x - a, k = y - b$, then Taylor's theorem becomes
$$f(x,y) = f(a,b) + \left((x-a)\frac{\partial}{\partial x} + (y-b)\frac{\partial}{\partial y}\right) f(a,b) + \frac{1}{2!}\left((x-a)\frac{\partial}{\partial x} + (y-b)\frac{\partial}{\partial y}\right)^2 f(a,b)$$
$$+ \cdots + \frac{1}{(n-1)!}\left((x-a)\frac{\partial}{\partial x} + (y-b)\frac{\partial}{\partial y}\right)^{n-1} f(a,b)$$
$$+ \frac{1}{n!}\left((x-a)\frac{\partial}{\partial x} + (y-b)\frac{\partial}{\partial y}\right)^n f(a(1-\theta) + \theta x, b(1-\theta) + \theta y)$$
$$0 < \theta < 1.$$
Here (a,b) is called the centre of the expansion.
2. For $m = 1, 2, \dots n$, we can expand
$$\left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)^m = h\frac{\partial^m}{\partial x^m} + {}^mC_1\frac{\partial^m}{\partial x^{m-1}\partial y} + {}^mC_2\frac{\partial^m}{\partial x^{m-2}\partial y^2} + \cdots + {}^mC_m\frac{\partial^m}{\partial y^m}$$
3. For $a = 0 = b$, we get the Maclaurin's development of the Taylor's theorem.

   If $f(x,y)$possesses continuous n^{th} order partial derivatives in the neighbourhood of the origin, then

$$f(x,y) = f(0,0) + \left(x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y}\right)f(0,0) + \frac{1}{2!}\left(x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y}\right)^2 f(0,0) + \cdots$$
$$+ \frac{1}{(n-1)!}\left(x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y}\right)^{n-1} f(0,0) + \frac{1}{n!}\left(x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y}\right)^{n} f(\theta x, \theta y)$$
$$0 < \theta < 1.$$

Problems

1. Expand $f(x,y) = e^x \log(1+y)$ upto third degree terms in x and y.
   Solution: Since $e^x \log(1+y)$ possesses continuous partial derivatives of order n in the neighbourhood of the origin, we can expand $e^x \log(1+y)$, and we have $f(0,0) = e^0 \log(1+0) = 0$.

$$f_x = e^x \log(1+y), \qquad f_x(0,0) = 0$$
$$f_y = \frac{e^x}{1+y}, \qquad f_y(0,0) = 1$$
$$f_{x^2} = e^x \log(1+y), \qquad f_{x^2}(0,0) = 0$$
$$f_{xy} = \frac{e^x}{1+y}, \qquad f_y(0,0) = 1$$
$$f_{y^2} = -\frac{e^x}{(1+y)^2}, \qquad f_{y^2}(0,0) = -1$$
$$f_{x^3} = e^x \log(1+y), \qquad f_{x^3}(0,0) = 0$$
$$f_{x^2y} = \frac{e^x}{1+y}, \qquad f_{x^2y}(0,0) = 1$$
$$f_{xy^2} = -\frac{e^x}{(1+y)^2}, \qquad f_{xy^2}(0,0) = -1$$
$$f_{y^3} = \frac{2e^x}{(1+y)^3}, \qquad f_{y^3}(0,0) = 2$$
$$\therefore f(x,y) = f(0,0) + \left(x\frac{\partial f}{\partial x}(0,0) + y\frac{\partial f}{\partial y}(0,0)\right) + \frac{1}{2!}(x^2\frac{\partial^2 f}{\partial x^2}(0,0) + 2xy\frac{\partial^2 f}{\partial x \partial y}(0,0)$$
$$+ y^2\frac{\partial^2 f}{\partial y^2}(0,0) + \cdots$$
$$\therefore e^x \log(1+y) = 0 + (x \cdot 0 + y \cdot 1) + \frac{1}{2!}\left(x^2 \cdot 0 + 2xy \cdot 1 + y^2(-1)\right) + \frac{1}{3!}(x^3 \cdot 0$$
$$+ 3x^2y(1) + 3xy^2(-1) + y^3(2))$$
$$\boldsymbol{e^x \log(1+y) = y + \frac{1}{2!}\left(2xy - y^2\right) + \frac{1}{6}\left(3x^2y - 3xy^2 + 2y^3\right) + \cdots}$$

2. Expand $f(x,y) = \sin xy$ upto second degree terms with centre at $(1, \pi/2)$.
   Solution:

$$f(1, \frac{\pi}{2}) = 1$$
$$f_x = y\cos xy, \qquad f_x\left(1, \frac{\pi}{2}\right) = \frac{\pi}{2}\cos\frac{\pi}{2} = 0$$
$$f_y = x\cos xy, \qquad f_y\left(1, \frac{\pi}{2}\right) = 0$$

$$f_{x^2} = -y^2 \sin xy, \qquad f_{x^2}\left(1,\frac{\pi}{2}\right) = -\frac{\pi^2}{4}$$

$$f_{xy} = \cos xy - xy \sin xy, \qquad f_{xy}\left(1,\frac{\pi}{2}\right) = -\frac{\pi}{2}$$

$$f_{y^2} = -x^2 \sin xy, \qquad f_{y^2}\left(1,\frac{\pi}{2}\right) = -1$$

$$\therefore f(x,y) = f\left(1,\frac{\pi}{2}\right) + \left((x-1)f_x(1,2) + \left(y-\frac{\pi}{2}\right)f_y\left(1,\frac{\pi}{2}\right)\right) + \frac{1}{2!}((x-1)^2 f_{x^2}(1,2)$$

$$+ 2(x-1)\left(y-\frac{\pi}{2}\right)f_{xy}\left(1,\frac{\pi}{2}\right) + \left(y-\frac{\pi}{2}\right)^2 f_{y^2}\left(1,\frac{\pi}{2}\right) + \cdots)$$

$$= 1 + \left((x-1)\cdot 0 + \left(y-\frac{\pi}{2}\right)\cdot 0\right) + \frac{1}{2!}((x-1)^2\left(-\frac{\pi^2}{4}\right) + 2(x-1)\left(y-\frac{\pi}{2}\right)\left(\frac{\pi}{2}\right)$$

$$+ \left(y-\frac{\pi}{2}\right)^2(-1) + \cdots$$

$$\boldsymbol{\sin xy = 1 - \frac{\pi^2}{8}(x-1)^2 + \frac{\pi}{2}(x-1)\left(y-\frac{\pi}{2}\right) - \frac{1}{2}\left(y-\frac{\pi}{2}\right)^2 + \cdots}$$

3. If $f(x,y)$ possesses continuous partial derivatives of certain higher order, given $f(1,2) = 10.0$, $f(1.01, 2.01) = 10.27$ and $f(0.99, 2.02) = 10.17$, approximate $f_x(1,2)$ and $f_y(1,2)$.
   Solution: By Taylor's theorem,
   $f(a+h, b+k) = f(a,b) + \left(hf_x(a,b) + kf_y(a,b)\right)$ + 2nd and higher degree terms in h and k.
   If h and k are small, the 2nd and higher degree terms in h and k can be neglected, and hence
   $$f(a+h, b+k) - f(a,b) \cong hf_x(a,b) + kf_y(a,b).$$
   For $a = 1, b = 2, h = k = 0.1$, we get
   $$f(1.01, 2.01) = f(1,2) = (0.1)f_x(1,2) + (0.1)f_y(1,2)$$
   $$\therefore f_x(1,2) + f_y(1,2) = \frac{1}{0.1}\left(f(1.01,2.01) - f(1,2)\right) = \frac{1}{0.1}(10.27 - 10.07) = 20$$
   ... **(1)**
   For $h = -0.01, k = 0.02, a = 1, b = 2,$
   $$f(0.99, 2.02) - f(1,2) = -0.01f_x(1,2) + 0.02f_y(1,2)$$
   $$\therefore f_x(1,2) + 2f_y(1,2) = 10$$
   ... **(2)**
   Solving (1) and (2), we get
   $$\boldsymbol{f_x(1,2) = 10 = f_y(1,2).}$$

Exercises

1. Expand $f(x,y) = e^x \cos y$ about $(0,0)$ upto 3rd degree terms.
2. Expand $f(x,y) = \tan^{-1}(y/x)$ about $(1,1)$ upto second degree terms.
3. Expand $f(x,y) = (1 + x + y^2)^{1/2}$ about $(1,0)$ upto second degree terms.
4. Expand $f(x,y) = e^{x/y}$ about $(1,1)$ upto third degree terms.
5. Expand $f(x,y) = \log(1 + x + y)$ about $(0,0)$ upto third degree terms.
6. If $f(x,y) = \tan^{-1}(xy)$, compute $f(0.9, -1.2)$ approximately.
7. If $f(x,y)$ is such that $f(0.2,0.2) = 2.72$, $f_x(0.2, 0.2) = 8.16$, and $f(0.21, 0.21) = 2.85$, approximate $f_y(0.2,0.2)$.